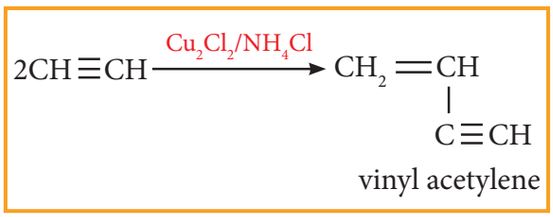
4. Polymerisation:

Alkyne undergoes two types of polymerisation reaction

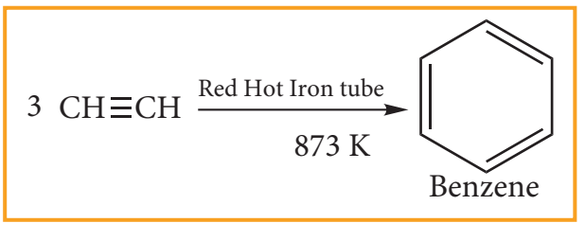
(i) Linear Polymerisation:

Ethyne forms linear polymer, when passed into a solution of cuprous chloride and ammonium chloride.



(ii) Cyclic Polymerisation:

Ethyne undergoes cyclic polymerization on passing through red hot iron tube. Three molecules of ethynepolymerises to benzene.



13.4.4 Uses of Alkynes

1) Acetylene is used in oxy acetylene torch used for welding and cutting metals.

2) It is used for manufacture of PVC, polyvinyl acetate, polyvinyl ether, orlon and neoprene rubbers.

Evaluate Yourself

20) $CH \equiv C - CH_3 + H_2 \xrightarrow{Pt} ? \xrightarrow{H_2} ?$

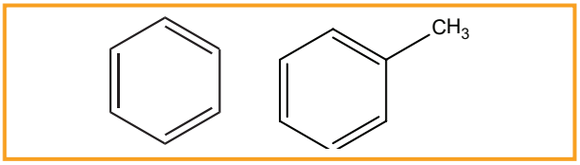
13.5. Aromatic Hydrocarbons

Take a moment and think of substances that have a strong fragrance. What kind of things come to your mind?

Perfume, Vanila or cinnamon? They smell differently, they have something in common. These substances are made of aromatic compounds [Greek: Aroma-Pleasant smelling]. However, some compounds are chemically aromatic but do not have distinct smell. The aromatic hydrocarbons are classified depending upon number of rings present in it.

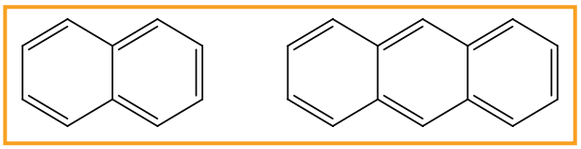
(i) Monocyclic aromatic hydrocarbon (MAH)

(Ex) Benzene (C_6H_6) and Toluene (C_7H_8)



(ii) Polycyclic aromatic hydrocarbon(PAH)

(Ex) Naphthalene ($C_{10}H_8$) and Anthracene ($C_{14}H_{10}$)



Evaluate Yourself

21) Calculate the number of rings present in $C_{18}H_{12}$.

13.5.1. Nomenclature and Isomerism

- We have already discussed about nomenclature of aromatic hydrocarbons in Unit:11. The first member of aromatic hydrocarbon is benzene (C_6H_6)

Unit 13 - jagan.indd 204 22-01-2021 12:51:38